Abuzar Mahmood — Curriculum Vitae

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Education

Ph.D., Neuroscience, Brandeis University, Waltham, MA — 2023 - GPA: 3.98 - Thesis Advisor: Donald B. Katz - Thesis Title: *Dynamic Neural Responses and Network Interactions during Taste Processing*

M.S., Neuroscience, Brandeis University, Waltham, MA — 2019 - GPA: 3.96

B.S., Physics, University of Missouri, Columbia, MO — 2017 - GPA: 3.95 (Summa Cum Laude) - Minors in Mathematics, Chemistry, Biology, and Computational Neuroscience

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Employment History

Postdoctoral Associate, Whitehead Institute, MIT, Cambridge, MA — July 2026 – Present

Postdoctoral Research Fellow, Tufts Medical Center, MA — July 2023 – Present - Project: Bioinformatics analysis of the effect of AT2R activation on cardiomyopathy in obesity and diabetes

Visiting Research Scientist, Brandeis University, MA — Mar 2026 – Present

Postdoctoral Fellow, Brandeis University, MA — Nov 2023 – Mar 2026 - Swartz Foundation Computational Neuroscience Postdoctoral Fellow (2023–25) - Project: Dynamics of information flow during the evoked taste response

Postdoctoral Associate, Brandeis University, MA — Mar 2023 – Nov 2023 - Project: Dynamics of information flow during the evoked taste response

Ph.D. Researcher, Brandeis University, MA — 2017 – Feb 2023 - Thesis Advisor: Donald B. Katz - Thesis Title: *Multi-region Coordination for Taste Processing in the Rodent Brain* - Investigated dynamics of coordination between the basolateral amygdala and gustatory cortex; tested whether the regions phenomenologically behave as an attractor network

Data Science Intern, Trackstar, NY — 2022 - Developed computational models to detect outliers and changepoints in online timeseries data; contributed to infrastructure for online inference

Undergraduate Researcher, University of Missouri, MO — 2013–2017 - Research Advisor: Lakshmi Pulakat - Projects investigating novel pharmaceutical drugs for treatment of cardiovascular disease and diabetes

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Publications

0.3.1 Neuroscience

1. Mazzio, C.T., Flashman, S., Mahmood, A., Baas-Thomas, N., Lin, J.Y., Komar, K. (2026). Stimulus-Specific and Generalized Taste Aversion Behaviors and Their Relationship to Cortical Dynamics. *bioRxiv* 2026.06.02.729621.
2. Baas-Thomas, N.*, Mahmood, A.*, Mukherjee, N., Maigler, K.C., Wang, Y., Katz, D.B. (2026). The Ingestive Response Reflects Neural Dynamics in Gustatory Cortex. *eLife* 15:RP110977. *Co-first authors
3. Mukadam, N., Mahmood, A., Cronin-Golomb, A., DeGutis, J. (2026). Subjective Cognitive Concerns and Global Metacognitive Bias in Prodromal Parkinson's and Parkinson's Disease without Cognitive Impairment. *Neuropsychology*. Epub ahead of print. PMID: 41989436. [Press Release]

4. Mahmood, A., Steindler, J., Katz, D.B. (2026). Sensory and Palatability Coding of Taste Stimuli in Cortex Involves Dynamic and Asymmetric Cortico-Amygdalar Interactions. *Journal of Neurophysiology*.
5. Mahmood, A. (2025). pytau: A Python Package for Streamlined Changepoint Model Analysis in Neuroscience. *Journal of Open Source Software*, 11(117), 8509.
6. Mahmood, A., Steindler, J., Germaine, H., Katz, D.B. (2023). Coupled Dynamics of Stimulus-Evoked Gustatory Cortical and Basolateral Amygdalar Activity. *Journal of Neuroscience*, 43(3), 386–404.
7. Stone, B.T., Lin, J.-Y., Mahmood, A., Sanford, A.J., and Katz, D.B. (2022). LiCl-Induced Sickness Modulates Rat Gustatory Cortical Responses. *PLOS Biology* 20, e3001537.
8. Lin, J.-Y., Stone, B.T., Herzog, L.E., Nanu, R., Mahmood, A., and Katz, D.B. (2021). The Function of Groups of Neurons Changes from Moment to Moment. *Current Opinion in Physiology* 20, 1–7.

0.3.2 Cardiovascular Disease and Diabetes

1. Belenchia AM, Boukhalfa A, DeMarco VG, Mehm A, Mahmood A, Liu P, et al. (2023). Cardiovascular Protective Effects of NP-6A4, a Drug with the FDA Designation for Pediatric Cardiomyopathy, in Female Rats with Obesity and Pre-Diabetes. *Cells* 12(10):1373.
2. Gavini, M.P.*, Mahmood, A.*, Belenchia, A.M., Beauparlant, P., Kumar, S.A., Ardhanari, S., et al. (2021). Suppression of Inflammatory Cardiac Cytokine Network in Rats with Untreated Obesity and Pre-Diabetes by AT2 Receptor Agonist NP-6A4. *Frontiers in Pharmacology* 12. *Co-first authors
3. Lum-Naihe, K., Toedebusch, R., Mahmood, A., Bajwa, J., Carmack, T., Kumar, S.A., et al. (2017). Cardiovascular Disease Progression in Female Zucker Diabetic Fatty Rats Occurs via Unique Mechanisms Compared to Males. *Sci Rep* 7, 17823.
4. Luck, C., DeMarco, V.G., Mahmood, A., Gavini, M.P., and Pulakat, L. (2017). Differential Regulation of Cardiac Function and Intracardiac Cytokines by Rapamycin in Healthy and Diabetic Rats. *Oxidative Medicine and Cellular Longevity* 2017, 1–17.
5. Arnold, N., Mahmood, A., Ramdas, M., Ehlinger, P.P., and Pulakat, L. (2017). Regulation of the Cardioprotective Adiponectin and Its Receptor AdipoR1 by Salt. *Can. J. Physiol. Pharmacol.* 95, 305–309.
6. Gul, R., Mahmood, A., Luck, C., Lum-Naihe, K., Alfadda, A.A., Speth, R.C., and Pulakat, L. (2015). Regulation of Cardiac miR-208a, an Inducer of Obesity, by Rapamycin and Nebivolol. *Obesity* 23, 2251–2259.
7. Mahmood, A., and Pulakat, L. (2015). Differential Effects of -Blockers, Angiotensin II Receptor Blockers, and a Novel AT2R Agonist NP-6A4 on Stress Response of Nutrient-Starved Cardiovascular Cells. *PLOS ONE* 10, e0144824.

In-Progress Manuscripts

0.4.1 Neuroscience

1. Rohrer, D.M., Nobili, C.C.C.M, Jiang, W., Mahmood, A., Knight, R.A., Gutsell, J., Katz, D.B. Am I Disgust? Exploring Disgust Responses Across Stimuli.
2. Mahmood, A.*, Calia-Bogan, V.*, Steindler, J., Katz, D.B., Miller, P. Characterizing Intra-State Dynamics of Gustatory Cortical Taste-Evoked Activity. *Equal contributors

0.4.2 Cardiovascular Disease and Diabetes

1. Mahmood, A., *Mehm, A.*, Mooney, B., DeMarco, V.G., Pulakat, L. Pathology and Signaling networks of cardio-pulmonary axis in a new female model for HFpEF syndrome: Uncovering new drug targets. *Equal contributors
2. Chaudhary, P.K., Mahmood, A., Valencia, D., Reyelt, L., Hines, I., Zhang, Y., et al. Development of a Rodent Stage-4 Pressure Ulcer Model.
3. Mahmood, A., Kiet Duong A., Mansour, A., Pulakat, L. CNN-assisted semi-automated histological evaluation of interstitial and perivascular fibrosis.

Published Abstracts

1. Mansour A, Herlihy M, Mahmood A, et al. Abstract 1880 Preventing Colistin Nephrotoxicity Caused by Systemic Delivery via Localized, Needle-Free Transdermal Delivery of Colistin Methanesulfonate Micromist: A Paradigm Shift in Antibiotic Treatment. *Journal of Biological Chemistry*, vol. 301, no. 5, May 2025, p. 109045. DOI.org (Crossref), https://doi.org/10.1016/j.jbc.2025.109045.
2. Pulakat L, DeMarco VG, Mahmood A, Mehm A, Tang Y, Martin GL, et al. Proteomics of the Cardiopulmonary Pathology in an Obese and Diabetic Female Rat Model – Evidence for a New Female HFpEF Rat Model. *9th International Caparica Conference on Analytical Proteomics*, Lisbon, Portugal (2024)
3. Pulakat L, Mahmood A, Belenchia A, Gavini M, Liu Pi, Mooney B, et al. Abstract 14318: Sex Differences in Treatment Responses to NP-6A4, a Cardiomyopathy Drug With the FDA Designation, in Heart Disease Induced by Untreated Obesity. *Circulation* 144(Suppl_1):A14318 (2021)
4. Gavini MP, Mahmood A, Belenchia AM, Beauparlant P, Kumar S, Ardhanari S, et al. Abstract 16430: Activation of AT2 Receptor by NP-6A4 Improves Cardiac Function by Inducing a Novel Cardio-Reparative Protein-Micro RNA Network in Zucker Obese Rats With Untreated Obesity and Diabetes. *Circulation* 140(Suppl_1):A16430 (2019)
5. Toedebusch R, Lum-Naihe K, Mahmood A, Bajwa J, Kumar S, Ardhanari S, et al. Mechanistic Insights into Diabetes and the Progression of Cardiovascular Disease in Female Rats. *The FASEB Journal* 31(S1):1014.17 (2017)
6. Pulakat L, Gavini M, Mahmood A, Belechia A, Toedebusch R, Beauparlant P, et al. Abstract P353: Attenuation of Cardiac Fibrosis, Hypertrophy and Myopathy by AT2R Agonist NP-6A4. *Hypertension* 70(suppl_1):AP353 (2017)
7. Mahmood A, Raja A, Pulakat L. Droplette – A Fluid Dynamics Driven Platform for Transdermal and Intra-Cellular Delivery of Large Molecules. *The FASEB Journal* 31(S1):924.7 (2017)
8. Belenchia AM, Beauparlant P, Mahmood A, Bajwa J, Zhang Q, Khare S, et al. Cardiovascular Protective vs. Anti-Cancer Properties: Novel Actions of the AT2R Agonist, NP-6A4. *The FASEB Journal* 31(S1):lb680 (2017)
9. Beauparlant P, Mahmood A, Toedebusch R, DeMarco V, Ardhanari S, Kumar S, et al. Cardiovascular Protective Effects of AT2R Activation by Peptide Drug NP-6A4. *The FASEB Journal* 31(S1):688.10 (2017)
10. Mahmood A, Pulakat L. Abstract P623: Nutrient Stress Response of Cardiovascular Cells to -Blockers, ARB and AT2R Agonists. *Hypertension* 66(suppl_1):AP623 (2015)
11. Mahmood A, Gul R, Luck C, Pulakat L. Role of Cardiac mir-208a in Nebivolol-Mediated Signaling. *The FASEB Journal* 29(S1):716.15 (2015)
12. Lum-Naihe K, Raja A, Bajwa J, Mahmood A, Luck C, Emter C, et al. Sex Differences in the Progression of Diabetes-Associated Cardiac Pathology. *The FASEB Journal* 29(S1):964.9 (2015)
13. Lum-Naihe K, Mahmood A, Bajwa J, Emter CA, Pulakat L. Abstract P636: Sex Differences

in Cardioprotective AT2R Expression in Diabetic Rats and Its Correlation with Myocardial Damage. *Hypertension* 66(suppl_1):AP636 (2015)

Poster Presentations

1. Mahmood A., Mansour A., Duong K.A., Pulakat L. Machine Learning-Assisted Semi-Automation of Interstitial Fibrosis Measurement in Cardiac Sections. *MCRI Retreat 2025*, Woods Hole — Honorable Mention
2. Mahmood A., DeMarco V.G., Mehm A., Tang Y., Martin G.L., Agrawal S., Mooney B., Pulakat L. Impact of NP-6A4 on the pathology of heart-lung continuum in a female model for metabolic syndrome and heart failure with preserved ejection fraction. *2024 Angiotensin Conference GRC*, Lucca, Italy (2024)
3. Mahmood A., Steindler J., Katz D.B. Basolateral Amygdala and Gustatory Cortex Interact Bidirectionally During Taste Processing in Rodents. *Society for Neuroscience Annual Meeting* (November 2024)
4. Mahmood A., Mehm A., Liu P., Mooney B., Gavini M.P., Tang Y., and Pulakat L. Characterizing Pathology and Signaling in Obese and Diabetic Female Rats with Heart Disease as a Model for HFpEF. *MCRI Annual Retreat*, Woods Hole, MA (September 2024)
5. Mahmood A., Steindler J., Katz D.B. Basolateral Amygdala and Gustatory Cortex Interact Bidirectionally During Taste Processing in Rodents. *Society for Neuroscience Annual Meeting* (November 2023) — Attendance supported by Society for Neuroscience Trainee Professional Development Award
6. Mahmood A., Steindler J., Stone B.T., Katz D.B. Dynamic Coupling of Taste-Evoked Gustatory Cortical and Basolateral Amygdalar Activity. *ACHEMS XLV*, FL (April 2023)
7. Mahmood A., Steindler J., Stone B.T., Katz D.B. Gustatory Cortex and Basolateral Amygdala Communication in Innate Taste Processing.
   - *International Symposium on Olfaction and Taste*, Portland, WA (August 2020)
   - First Place, Graduate Poster Competition, Volen Center for Complex Systems Retreat (Sept 2021)
8. Mahmood A., Luck C., DeMarco V.G., Gavini M., and Pulakat L. Differential Regulation of Cardiac Function and Intracardiac Cytokines by Rapamycin in Healthy and Diabetic Rats. *University of Missouri Cardiovascular Day XXIV Poster Competition* (Feb 2017) — First Place, Undergraduate Competition
9. Mahmood A., Gavini M., Senthilkumar A., Carmack T., DeMarco V., and Pulakat L. Investigating the role of the Angiotensin II Type 2 Receptor in Protective Cardiac Remodeling. *University of Missouri Summer Forum* (July 2016)
10. Mahmood A., Raja A., Gavini M., and Pulakat L. Droplette — A Novel Method of Low-Pressure Transdermal Delivery to Chronic Wounds. *Missouri Life Sciences Week* (Apr 2016) — First Place, Undergraduate Competition, Life Science and Biomedical Engineering Technologies and Informatics category
11. Mahmood A., and Pulakat L. Improved Survival of Nutrient-Starved Human and Mouse Cardiovascular Cells by A Novel AT2 Receptor Agonist NP-6A4. - *Nutrition and Exercise Physiology Corporate Affiliate Board Meeting* (Nov 2015) - *University of Missouri Health Sciences Research Day* (Nov 2015)
12. Mahmood A., and Pulakat L. Nutrient Stress Response of Cardiovascular Cells to -Blockers, ARB and AT2R Agonists. - *University of Missouri Undergraduate Research and Creative Achievements Summer Forum* (July 2015) - *Harry S. Truman VA Research Week Poster Competition* (Sept 2015) - *Hypertension 2015 Scientific Sessions* (Sept 2015)
13. Mahmood A., Gul R., Luck C., and Pulakat L. Role of Cardiac mir-208a in Nebivolol-Mediated Signaling. *Experimental Biology*, Boston (March 2015)

14. Mahmood A., and Pulakat L. Differential Effects of -Blockers and a Novel AT2R Agonist on Cardiovascular Cells. - Awarded Distinction, *University of Missouri Cardiovascular Day XXII Poster Competition* (Feb 2015) - *Missouri Life Sciences Week Poster Competition* (April 2015) - *University of Missouri Undergraduate Research and Creative Achievements Spring Forum* (April 2015)
15. Mahmood A., and Pulakat L. Electrical Impedance-Based Measurement of Cardiomyocyte Response to Cardio-protective Drugs. *University of Missouri Health Sciences Research Day* (Nov 2014)
16. Mahmood A., Gul R., and Pulakat L. Modulation of Electrical Impedance of Cardiomyocytes by Nebivolol and Novel AT2 Receptor Agonists. - *Harry S. Truman VA Research Week Poster Competition* (May 2014) - *Missouri Life Sciences Week Poster Competition* (April 2014)

0.6.1 Co-Authored Poster Presentations

1. Pulakat L., Chaudhary P.K., Chen H.H., Mahmood A., Valencia D., Hines I., Reyelt L., Zhang Y., Gavini M.P. Effects of needle-free therapeutic micromist carrying Tumor Necrosis Factor silencer RNA on expediting chronic wound healing, repairing deep muscle damage and improving survival in a new rat model. Abstract ID: 4452. *ASBMB Meeting 2026*, Washington DC (March 2026)
2. Mansour A., Herlihy M., Mahmood A., Velasquez F., Duong K., Hines I., Martin G., Chaudhary P.K., Gavini M.P., Sun H., Pulakat L. Preventing Colistin Nephrotoxicity Caused by Systemic Delivery via Localized, Needle-Free Transdermal Delivery of Colistin Methanesulfonate Micromist. *MCRI Retreat 2025*, Woods Hole, MA
3. Chaudhary P.K., Mahmood A., Nasencia D., Reyelt L., Hines I., Zhang Y., Duong A., Martin G., Gavini M.P. A Novel siRNA Therapy Delivered via Needle-Free Therapeutic Micromist to Expedite Chronic Wound Healing and Improve Survival. *MCRI Retreat 2025*, Woods Hole, MA
4. Baas-Thomas N., Mahmood A., Maigler K., Wang Y., and Katz D.B. The Ingestive Response Reflects Neural Dynamics in the Gustatory Cortex. *ACHEMS*, FL (April 2025)
5. Gray T.R., Mahmood A., Craddock A.E., White A.S., Goldstein I., Katz D.B. Gustatory-Olfactory Cortical Interactions in Response to Multimodal Stimuli. *ACHEMS*, FL (April 2025)
6. Pulakat L., Mahmood A., Mansour A., Chen H., Valencia D., Zhang Y., Gavini M. A Novel Preclinical Model for Obesity-Associated Stage 4 Pressure Ulcer and A Promising RNA-Based Therapeutic Approach. *Military Health System Research Symposium*, FL (Aug 2025)
7. Xu C., Mahmood A., Mansour A., and Pulakat L. Evaluation of the effect of human mesenchymal stem cells on diabetic foot wound in a rat model with severe diabetes. *Tufts Myocardial Research Institute Annual Retreat*, Woods Hole, MA (September 2024)
8. Calia-Bogan V., Mahmood A., Steindler J.R., Katz D.B. Taste-Evoked Intra-Epoch Dynamics in Gustatory Cortex.
    - *ACHEMS*, FL (April 2025)
    - *Society for Neuroscience Annual Meeting* (November 2024)
    - *Brandeis Sci-Fest*, Waltham, MA (Aug 2024)
9. Pulakat L., DeMarco V.G., Mahmood A., et al. Cardiac and Extracardiac Pathology of a Preclinical Model for Female-Specific HFpEF Syndrome. *AHA Hypertension Scientific Session*, Chicago, IL (Sept 2024)
10. Baas-Thomas N., Mahmood A., Wang Y., Katz D.B. Investigating the Neural Signals Driving the Consummatory Response in Rats.
    - *Society for Neuroscience Annual Meeting* (November 2024)
    - *International Symposium on Olfaction and Taste* (June 2024)
11. Mazzio C., Lin J.Y., Germaine H., Mahmood A., Katz D.B. Cortical Population Dynamics

Underlying Learned and Unlearned Aversive Behavior. *ACHEMS XLV*, FL (April 2023)

12. Pulakat L., Mahmood A., Belenchia A., Gavini M., Liu P., Mooney B., Tang Y., Mehm A., and Demarco V.G. Sex Differences in Treatment Responses to NP-6A4, a Cardiomyopathy Drug With the FDA Designation, in Heart Disease Induced by Untreated Obesity. *American Heart Association Meeting* (Nov 2021)
13. Maigler K.M., Stone B.T., Mahmood A., Lin J.Y., and Katz D.B. The Contribution of Lateral Hypothalamus to Cortical Palatability Coding. *Association for Chemoreception Science Meeting* (April 2021)
14. Stone B.T., Mahmood A., Lin J.Y., and Katz D.B. Representation of Illness and Its Functional Impact on Gustatory Processing. *International Symposium on Olfaction and Taste*, Portland (August 2020)
15. Gavini M.P., Mahmood A., Belenchia A.M., Beauparlant P., Kumar S.A., Ardhanari S., DeMarco V.G., and Pulakat L. Activation of AT2 Receptor by NP-6A4 Improves Cardiac Function by Inducing a Novel Cardio-Reparative Protein-Micro RNA Network in Zucker Obese Rats With Untreated Obesity and Diabetes. *American Heart Association Meeting* (Nov 2019)
16. Pulakat L., Gavini M.P., Mahmood A., Belenchia A.M., Beauparlant P., Kumar S.A., Ardhanari S., and DeMarco V.G. Attenuation of Cardiac Fibrosis, Hypertrophy and Myopathy by AT2R Agonist NP-6A4. *AHA Hypertension Meeting* (Sept 2017)
17. Belenchia A.M., Beauparlant P., Mahmood A., Bajwa J., Zhang Q., Khare S., and Pulakat L. Cardiovascular Protective vs. Anti-Cancer Properties: Novel Actions of the AT2R Agonist, NP-6A4. *Experimental Biology Meeting* (April 2017)
18. Toedebusch R., Lum-Naihe K., Mahmood A., Bajwa J., Kumar S., Ardhanari S., Demarco V., and Pulakat L. Mechanistic Insights into Diabetes and the Progression of Cardiovascular Disease in Female Rats. *Experimental Biology Meeting* (April 2017)
19. Beauparlant P., Mahmood A., Toedebusch R., Kumar S., Ardhanari S., Demarco V., and Pulakat L. Cardiovascular Protective Effects of AT2R Activation by Peptide Drug NP-6A4. *Experimental Biology Meeting* (April 2017)
20. Lum-Naihe K., Toedebusch R., Mahmood A., Bajwa J., Carmack T., Kumar S.A., Ardhanari S., DeMarco V.G., Emter C.A., and Pulakat L. Sex Differences in the Expression of Cardiac miR-29 Family microRNAs in Diabetic Male and Female Rats and Its Correlation with Increased Risk for Cardiac Damage in Diabetic Females. *Missouri Life Sciences Week* (April 2016) — Honorable Mention, Graduate Competition, Molecular and Cellular Biology
21. Bajwa J., Mahmood A., Gavini M., Pulakat L. Regulation of Neuroprotective Myeloid Cell Leukemia 1 by Rapamycin and AT2R Agonists in Dopaminergic Neuronal Cell Line. *Hypertension 2015 Scientific Sessions*, Washington DC (Sept 2015)
22. Lum-Naihe K., Raja A., Bajwa J., Mahmood A., Luck C., Emter C., Pulakat L. Sex Differences in the Progression of Diabetes-associated Cardiac Pathology. *Experimental Biology*, Boston (March 2015)
23. Arnold N., Mahmood A., Ramdas M., and Pulakat L. Suppression of the Cardio-protective molecule Adiponectin via a High-Salt Diet: A Potentially Pivotal Mechanism of Atherosclerosis in Salt-Induced Hypertension & Heart Disease. *University of Missouri Health Sciences Research Day* (Nov 2014)

Talks

Title	Venue	Year
	NIMH Computational Neuroscience Journal Club	2026

Title	Venue	Year
	Hamilos Lab Meeting, Whitehead Institute, MIT	2026
The Cortico-Amygdalar Interaction Dynamics Underlying Taste Perception & Action	MaTRIX Laboratory Meeting, Georgia Tech	Nov 2025
	Mathematical Biology Seminar, Brandeis University	2025
	Insights Team Meeting, Appcast Inc.	2025
The Cortico-Amygdalar Interaction Dynamics Underlying Taste Perception & Action	Systems Neuroscience Journal Club, Harvard Medical School	April 2025
Basolateral Amygdala and Gustatory Cortex Interact Bidirectionally During Taste Processing in Rodents	Swartz Foundation Annual Meeting, WA	2024
Basolateral Amygdala and Gustatory Cortex Interact Bidirectionally During Taste Processing in Rodents	Brandeis Neuro Postdoc Symposium, Waltham, MA	2024
Polak Young Investigator Award Lecture	Association for Chemoreception Sciences Annual Meeting, FL	2024
The Only Constant Is Change: Bespoke Changepoint Modelling in PyMC	PyMCon Web Series	2023
An Intro to Bayesian Modelling and Probabilistic Programming	Brandeis University, MA	2022
Gustatory Cortex and Basolateral Amygdala Coordination in Taste Processing	Brandeis University	Mar 2021, Oct 2021
Gustatory Cortex and Basolateral Amygdala Communication in Innate Taste Processing	NIH Blueprint Joint Symposium on Computational Neuroscience	2021
Role of Gustatory Cortex and Basolateral Amygdala Communication in Taste Processing	Brandeis University	2020
The Nonlinear Population Dynamics of Cortical Taste Processing	Systems/Computational Neuroscience Journal Club, Brandeis University, MA	2019

Support

ACCESS-CI Compute Grant, MED250058 NSF (Co-PI) — 2025 - Project Title: “CNN-based semi-automated evaluation of fibrosis in histological cardiac sections” - In-kind support of compute allocation valued at approx. $40,000

Computational Neuroscience Postdoctoral Fellowship, Swartz Foundation — 2023–25

ACCESS-CI Compute Grant, BIO230103 NSF (PI) — 2023–25 - Project Title: “Computational Processing and Modeling of Neural Ensembles in Identifying the Nonlinear Dynamics of Taste Perception” - In-kind support of compute allocation valued at approx. $40,000

XSEDE Compute Research Award, NSF — 2019–22

Computational Neuroscience Training Fellowship, NIH — 2017–19

Academic Hardware Grant, NVIDIA — 2018 - Project Title: “Open-Source Neuro-Technology: Democratizing High-Performance Hardware and Analysis Pipelines for Systems Neuroscience”

Life Sciences Undergraduate Research Opportunity, University of Missouri — 2016 - Project Title: “Investigating the Role of the Angiotensin II Type 2 Receptor in Protective Cardiac Remodeling” - Project Advisor: Lakshmi Pulakat

Undergraduate Research Travel Award, Office of Undergraduate Research, University of Missouri — 2015 - Awarded to present at Experimental Biology, Boston

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Awards

Poster of Distinction, Honorary Mention, MCRI Annual Retreat Poster Competition — 2025

Polak Young Investigator Award, Association for Chemoreception Sciences — 2024

Trainee Professional Development Award, Society for Neuroscience — 2023

First Place, Poster Competition, Volen Center for Complex Systems Retreat — 2021

First Place, Undergraduate Poster Competition, Missouri LS Week — 2016, 2017

Distinction, University of Missouri Cardiovascular Day XXII Poster Competition — 2015

0.9.1 Academic Awards

Award for Academic Distinction, University of Missouri — 2017 - Awarded to 10 students out of approx. 23,000 every year

Summer Research Fellowship, University of Missouri — 2016

Clifford W. Tompson Scholarship, Department of Physics, University of Missouri — 2016–2017

Junior Scholarship Award, Honors College, University of Missouri — 2016

Curator’s Grant in Aid Award, International Center, University of Missouri — 2015–2017

Col. Arthur C. Allen Scholarship, College of Arts and Science, University of Missouri — 2015–2017

Dean’s List, College of Arts and Science, University of Missouri — 2013–2017

International Merit Scholarship, International Center, University of Missouri — 2013–2017

Rosemary Dishman Scholarship, Department of Physics, University of Missouri — 2015

Sophomore Scholarship Award, Honors College, University of Missouri — 2015

Newell S. Gingrich Undergraduate Scholarship, Department of Physics, University of Missouri — 2014

Paul E. Basye Scholarship, Department of Physics, University of Missouri — 2014

Teaching

Workshop Instructor, Introduction to Git and GitHub, Brandeis University — 2024, 2025

Workshop Instructor, Probabilistic Programming and Bayesian Modeling, Brandeis University — 2021, 2025

Guest Lecturer, Advanced Data Analysis (with Dr. Shantanu Jadhav), Brandeis University — 2021 - Unsupervised clustering and Hidden Markov Modeling - Probabilistic Programming and Bayesian Modeling - LLM agent pipelines

Teaching Assistant, Applied Statistical Computing in R (with Prof. Xiaodong Liu), Brandeis University — 2019

Guest Lecturer and Teaching Assistant, Computational Neuroscience (with Dr. Leandro Alonso), Brandeis University — 2019 - Detecting circuit structure and non-random features in a connectivity matrix - Principal Component Analysis

Tutor, Quantitative Skills Center, Brandeis University — 2019

Teaching Assistant, Data Analysis and Statistics Workshop (with Prof. Paul Miller), Brandeis University — 2018

Help Sessions Tutor, University of Missouri, Department of Physics — 2015–2016

Mentorship

- **MCRI, Tufts Medical Center, MA**
  - Alex Mansour, Undergraduate — Data Collection and Analysis (2024–25)
    - Project: *Preventing systemic colistin nephrotoxicity using localized, needle-free transdermal delivery*
    - Co-author on manuscript (In Preparation)
  - Kiet Duong, Undergraduate — Data Collection and Analysis (2024–25)
    - Project: *Assessment of Cardiac Function and Microvascular Structure in a Rat Model of Pressure Ulcer Injury*
    - Co-author on manuscript (In Preparation)
  - Charles Xu, MD student — Data Collection and Analysis (2024)
    - Project: *Effect of human mesenchymal stem cells on foot wound in a rat model with severe diabetes*
  - Abigail Wu, Summer Student (Research Science Institute) — Data Analysis (2024)
    - Summer Project Thesis: *Effects of rapamycin on cardiac health and signaling in rats fed a cafeteria diet*
  - Nancy Hassan, Summer Student (Research Science Institute) — Data Analysis (2024)
    - Summer Project Thesis: *Effects of chemo-toxicity on iCell cardiomyocytes*
- **Brandeis University, MA**
  - Vincent Calia-Bogan, Undergraduate — Computational Modeling and Analysis (2023–25)
    - Computational Neuroscience Training Grant Awardee
    - Undergraduate Thesis: *Investigating Intra-State Dynamics in Gustatory Cortex* (High Honors)
    - Co-first author on manuscript (In Preparation)
  - Vasudeva Krishnamurthy Bhat, PhD Rotation Student — Statistical Modelling and Data Analysis (2025)

    - Project: *Detecting changepoints in electrophysiological spectra*
  - Hannah Germaine, PhD Rotation Student — Modeling and Data Analysis (2022)
    - Project: *Statistical Validation of Transition Alignment in Multiple Neural Population*
    - Co-author on 1 published article
  - Victor Suarez, PhD Rotation Student — Data Analysis (2020)
    - Project: *Neural Correlates of Variable Behavioral Output to Aversive Tastants*
  - Jessica Steindler, Post-Bac — Hardware construction, surgery, and data collection (2019–21)
    - Project: *Dynamics of the BLA-GC Interaction During Taste Processing*
    - Co-author on 2 published articles and 1 manuscript (In Preparation)
  - Thomas Murdy, Undergraduate — Surgery, data collection, and analysis (2020–21)
    - Undergraduate Thesis: *Investigating the BLA-GC interaction in CTA acquisition* (High Honors)
- **University of Missouri, MO**
  - Paige Beauparlant, Undergraduate — Data collection (2017)
    - Project: *Cardiovascular Protective Effects of AT2R Activation by Peptide Drug NP-6A4*
  - Laura Perry, Undergraduate — Data collection (2017)
  - Jamal Bajwa, Undergraduate — Data collection and analysis (2016–17)
    - Project: *Regulation of Neuroprotective MCL1 by Rapamycin and AT2R Agonists in Dopaminergic Neurons*

Leadership and Service

Executive Committee, Brandeis University Postdocs Association — 2023–26

Selection Committee, Invited Postdoc Research Colloquium, Brandeis University — 2023–26

Ad-hoc Reviewer (with Prof. Donald Katz): Journal of Neuroscience Methods, Journal of Neuroscience, Applied Physics Reviews

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Professional Memberships

- Member, Association for Chemoreception Sciences, — 2020–2025
- Member, Society for Neuroscience, — 2022–2025
- Student member, Sigma Pi Sigma, University of Missouri, Columbia, MO — 2015–2017
- Student member, American Society for Biochemistry and Molecular Biology, — 2014–2015
- Student member, Society for Physics Students, University of Missouri, Columbia, MO — 2013–2017
- Student member, American Physical Society, — 2014–2016

Open-Source Contributions

Full list at github.com/abuzarmahmood

blech_clust — Spike-sorting library for multi-channel electrophysiological recordings featuring automated waveform classification and clustering, quality control, and data handling methods.

pytau — Python package for Bayesian changepoint modelling with multiple inference algorithms (MCMC/ADVI), Bayesian nonparametric priors, and streamlined batch processing and model tracking. Published in JOSS.

neuRecommend — XGBoost-based extracellular electrophysiology spike-waveform classifier.

Autoslide — Convolutional neural network-based semi-automated evaluation of perivascular fibrosis in histological sections.

Skills

- **Experimental**
  - Stereotactic rodent surgeries, chronic implantation of multielectrode bundles and optical fibers
- **Software**
  - Python, R, MATLAB, Linux
  - MySQL and PostgreSQL (pgAdmin4, psycopg2)
  - Computing cluster environments (Brandeis, XSEDE Jetstream, ACCESS-CI)
- **Modeling**
  - *Machine Learning:* Standard models for regression, classification, and clustering; dimensionality reduction; time-series models (ARIMA, Holt-Winters, HMMs, changepoint models); probabilistic modeling and Bayesian inference including nonparametric priors
  - *Statistics:* Frequentist techniques (parametric/non-parametric), Bayesian statistics (hierarchical models, MCMC), computational neuroscience models (point-process models, drift-diffusion model)
  - *Frameworks:* numpy, scipy, scikit-learn, TensorFlow/Keras, PyMC3, pandas, PySpark, Kubernetes, AWS EC2 and ECS, HPC
- **Formal Coursework**
  - Mathematical Statistics, University of Missouri (2017)
  - Statistical Machine Learning, Brandeis University (2018)
  - Computational Neuroscience, Brandeis University (2019)
  - Bayesian Inference and Computational Statistics, Brandeis University (2019)
  - Advanced Data Analysis and Modeling, Brandeis University (2020)
- **Certifications**
  - DeepLearning.AI TensorFlow Developer Specialization, Coursera (2021)
    - Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning
    - Convolutional Neural Networks in TensorFlow
    - Natural Language Processing in TensorFlow
    - Sequences, Time Series and Prediction
  - Advanced Visualizations in Python, Coursera (2022)
  - DeepLearning.AI Machine Learning in Production Specialization, Coursera (2022)
    - Introduction to Machine Learning in Production
    - Machine Learning Data Lifecycle in Production
    - Machine Learning Modeling Pipelines in Production
    - Deploying Machine Learning Models in Production